# STS 531: Mathematical Statistics I

**Date:**

# Homework IX

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May 23, 2026

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**Problem 1** Let $Y_1 < Y_2 < Y_3 < Y_4$ be the order statistics of a random sample of size $n = 4$ from a population with density function

$$f_X(x) = \begin{cases} 6x(1-x), & 0 \leq x \leq 1 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the following probabilities:

**(a)** $P\left(0.3 < Y_3 < 0.5\right)$

**(b)** $P\left(\frac{1}{3} < Y_2 < \frac{3}{4}\right)$

**Solution** **(a)** The population's cumulative distribution function is

$$F_X(x) = \int_0^x 6t(1-t)\ dt \tag{1}$$

$$= 3x^2 - 2x^3, \tag{2}$$

for $0 \leq x \leq 1$. For $x < 0$, $F(x) = 0$, and for $x > 1$, $F(x) = 1$. Thus, (by Theorem 5.4.4 from Casella and Berger) the density of $Y_3$ is

$$f_{Y_3}(x) = \frac{4!}{2!1!} f_X(x)[F_X(x)]^2[1 - F_X(x)]^1 \tag{3}$$

$$= 12\left(6x(1-x)[3x^2 - 2x^3]^2[1 - 3x^2 + 2x^3]\right), \tag{4}$$

for $0 \leq x \leq 1$. Using this density, the desired probability is

$$P\left(0.3 < Y_3 < 0.5\right) = \int_{0.3}^{0.5} 12\left(6x(1-x)[3x^2 - 2x^3]^2[1 - 3x^2 + 2x^3]\right)\ dx. \tag{5}$$

To simplify this integral, let $u = 3x^2 - 2x^3$. Then $du = 6x(1-x)\ dx$. With this transformation,

$$\int_{0.3}^{0.5} 12\left(6x(1-x)[3x^2 - 2x^3]^2[1 - 3x^2 + 2x^3]\right)\ dx = 12 \int_{0.216}^{0.5} u^2(1-u)\ du \tag{6}$$

$$= \left[4u^3 - 3u^4\right]_{0.216}^{0.5} \tag{7}$$

$$\approx 0.2787. \tag{8}$$

Therefore, $P\left(0.3 < Y_3 < 0.5\right) \approx 0.2787$.

**(b)** Again we use Theorem 5.4.4 to obtain the density of $Y_2$:

$$f_{Y_2}(x) = \frac{4!}{1!2!} f_X(x) \left[F_X(x)\right]^1 \left[1 - F_X(x)\right]^2 \tag{9}$$

$$= 12(6x(1-x)[3x^2 - 2x^3][1 - 3x^2 + 2x^3]^2), \tag{10}$$

for $0 \le x \le 1$. The desired probability is then

$$P\left(\frac{1}{3} < Y_2 < \frac{3}{4}\right) = \int_{\frac{1}{3}}^{\frac{3}{4}} 12(6x(1-x)[3x^2 - 2x^3][1 - 3x^2 + 2x^3]^2)\, dx. \tag{11}$$

Letting $u = 3x^2 - 2x^3$, we have

$$\int_{\frac{1}{3}}^{\frac{3}{4}} 12(6x(1-x)[3x^2 - 2x^3][1 - 3x^2 + 2x^3]^2)\, dx = 12 \int_{\frac{7}{27}}^{\frac{27}{32}} u(1-u)^2\, du \tag{12}$$

$$= 12 \int_{\frac{7}{27}}^{\frac{27}{32}} u[1 - 2u + u^2]\, du \tag{13}$$

$$= 12 \int_{\frac{7}{27}}^{\frac{27}{32}} u - 2u^2 + u^3\, du \tag{14}$$

$$= \left[6u^2 - 8u^3 + 3u^4\right]_{\frac{7}{27}}^{\frac{27}{32}} \tag{15}$$

$$\approx 0.7091. \tag{16}$$

Therefore, $P\left(\frac{1}{3} < Y_2 < \frac{3}{4}\right) \approx 0.7091$. ■

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**Problem 2** Let $X_1, X_2, X_3$ be independent random variables with probability density function

$$f(x) = \begin{cases} \exp\left\{-(x-\theta)\right\}, & x > \theta \\ 0 & \text{elsewhere.} \end{cases}$$

Determine $c = c(\theta)$ for which $P(\theta < Y_3 < c) = 0.90$, where $Y_3$ is the third order statistic.

**Solution** The cumulative distribution function for the population is

$$F_X(x) = \int_{\theta}^{x} \exp\left\{-(t-\theta)\right\}\, dt \tag{17}$$

$$= \exp(\theta) \int_{\theta}^{x} \exp(-t)\, dt \tag{18}$$

$$= -\exp(\theta)\left[\exp(-x) - \exp(-\theta)\right] \tag{19}$$

$$= 1 - \exp\{-(x-\theta)\}, \tag{20}$$

for $x > \theta$. By Theorem 5.4.4, we can find the probability density function of $Y_3$:

$$f_{Y_3}(x) = \frac{3!}{2!0!} f_X(x)[F_X(x)]^2 \tag{21}$$

$$= 3\exp\{-(x-\theta)\}[1-\exp\{-(x-\theta)\}]^2, \tag{22}$$

for $x > \theta$. Now,

$$P(\theta < Y_3 < c) = \int_\theta^c f_{Y_3}(x)\, dx \tag{23}$$

$$= \int_\theta^c 3\exp\{-(x-\theta)\}[1-\exp\{-(x-\theta)\}]^2\, dx. \tag{24}$$

Let $u = \exp\{-(x-\theta)\}$. With this transformation, we have

$$\int_\theta^c 3\exp\{-(x-\theta)\}[1-\exp\{-(x-\theta)\}]^2\, dx = -3 \int_1^{\exp(\theta-c)} (1-u)^2\, du. \tag{25}$$

Now let $v = 1-u$. Then

$$-3 \int_1^{\exp(\theta-c)} (1-u)^2\, du = \int_0^{1-\exp(\theta-c)} 3v^2\, dv \tag{26}$$

$$= \left[v^3\right]_0^{1-\exp(\theta-c)} \tag{27}$$

$$= (1-\exp(\theta-c))^3. \tag{28}$$

Setting this equal to 0.90 and solving for $c$, we have

$$c = \theta - \ln\left(1 - 0.9^{\frac{1}{3}}\right) \approx \theta + 3.366.$$

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**Problem 3** For the sample from Problem 2, find the pdf of $S_M$ (the sample median), where

$$S_M = \begin{cases} Y_{\frac{1}{2}(n+1)} & \text{if } n \text{ is odd} \\ \frac{1}{2}\left(Y_{\frac{n}{2}} + Y_{\frac{n}{2}+1}\right) & \text{if } n \text{ is even.} \end{cases}$$

**Solution** With a sample size of $n = 3$, we have

$$f_{S_M}(x) = f_{Y_2}(x) \tag{29}$$

$$= \frac{3!}{(2-1)!(3-2)!} f_X(x)[F_X(x)]^{2-1}[1-F_X(x)]^{3-2} \tag{30}$$

$$= 6\exp[-2(x-\theta)]\{1-\exp[-(x-\theta)]. \tag{31}$$

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**Problem 4** Suppose $X_1, X_2, X_3$ are three independent random variables having probability density function

$$f_X(x) = \begin{cases} 2(2-x), & 1 \le x \le 2 \\ 0, & \text{elsewhere.} \end{cases}$$

**(a)** Find the pdf of $S_M$.

**(b)** Calculate $P(Y_2 > m)$ where $m$ is the median of $f$.

**Solution** **(a)** The cumulative distribution function for the population is

$$F_X(x) = \int_1^x 2(2-t)\ dt \tag{32}$$
$$= [4t - t^2]_1^x \tag{33}$$
$$= 4x - x^2 - 3. \tag{34}$$

Since our sample size is $n = 3$, we have $S_M = Y_2$. Therefore, the pdf of $S_M$ is

$$f_{S_M}(x) = f_{Y_2}(x) \tag{35}$$
$$= \frac{3!}{(2-1)!(3-2)!} f_X(x)[F_X(x)]^{2-1}[1 - F_X(x)]^{3-2} \tag{36}$$
$$= 12(2-x)[4x - x^2 - 3][1 - (4x - x^2 - 3)]. \tag{37}$$

**(b)** The median of $f$ can be found by solving the equation

$$\int_1^m 2(2-x)\ dx = 0.5.$$

We have

$$\int_1^m 2(2-x)\ dx = 4m - m^2 - 3. \tag{38}$$

Setting this equal to 0.5 and solving for $m$ using the quadratic formula yields $m = 2 \pm \frac{\sqrt{2}}{2}$. But $2 + \frac{\sqrt{2}}{2}$ is outside the range for the random variable representing the population. Therefore, $m = 2 - \frac{\sqrt{2}}{2}$ and

$$P(Y_2 > m) = P\left(Y_2 > 2 - \frac{\sqrt{2}}{2}\right) \tag{39}$$
$$= 1 - F_X\left(2 - \frac{\sqrt{2}}{2}\right) \tag{40}$$
$$= 1 - 0.5 \tag{41}$$
$$= 0.5. \tag{42}$$

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**Problem 5** Let $X_1, X_2, \ldots X_n$ be independent random variables and suppose that $X_i$ has density $f_i$ $(i = 1, 2, \ldots n)$. Find the density of

**(a)** $Y_1 = \min(X_1, X_2, \ldots, X_n)$

**(b)** $Y_n = \max(X_1, X_2, \ldots, X_n)$

**Solution** **(a)** Then cumulative distribution function for $Y_1$ is

$$\begin{aligned}
F_{Y_1}(x) &= P(Y_1 \leq x) &(43)\\
&= 1 - P(Y_1 > x) &(44)\\
&= 1 - P(\min(X_1, X_2, \ldots, X_n) > x) &(45)\\
&= 1 - P(X_1 > x, X_2 > x, \ldots X_n > x). &(46)
\end{aligned}$$

Since $X_1, X_2, \ldots X_n$ are independent, we have

$$\begin{aligned}
1 - P(X_1 > x, X_2 > x, \ldots X_n > x) &= 1 - \prod_{i=1}^{n} P(X_i > x) &(47)\\
&= 1 - \prod_{i=1}^{n} [1 - P(X_i \leq x)] &(48)\\
&= 1 - \prod_{i=1}^{n} [1 - F_i(x)]. &(49)
\end{aligned}$$

So $F_{Y_1}(x) = 1 - \prod_{i=1}^{n} [1 - F_i(x)]$. Therefore,

$$\begin{aligned}
f_{Y_1}(x) &= -\frac{d}{dx} \prod_{i=1}^{n} [1 - F_i(x)] &(50)\\
&= -\left[ \frac{d}{dx}(1 - F_1(x)) \prod_{\substack{1 \leq i \leq n \\ i \neq 1}} [1 - F_i(x)] + \ldots + \frac{d}{dx}(1 - F_n(x)) \prod_{\substack{1 \leq i \leq n \\ i \neq n}} [1 - F_i(x)] \right] &(51)\\
&= f_1(x) \prod_{\substack{1 \leq i \leq n \\ i \neq 1}} [1 - F_i(x)] + \ldots + f_n(x) \prod_{\substack{1 \leq i \leq n \\ i \neq n}} [1 - F_i(x)] &(52)\\
&= \sum_{i=1}^{n} f_i(x) \prod_{\substack{1 \leq j \leq n \\ j \neq i}} [1 - F_j(x)]. &(53)
\end{aligned}$$

**(b)** The cumulative distribution function for $Y_n$ is

$$F_{Y_n}(x) = P(Y_n \le x) \tag{54}$$
$$= P(\max(X_1, X_2, \dots X_n) \le x) \tag{55}$$
$$= P(X_1 \le x, X_2 \le x, \dots X_n \le x) \tag{56}$$
$$= P(X_1 \le x)P(X_2 \le x) \cdots P(X_n \le x) \tag{57}$$
$$= \prod_{i=1}^{n} P(X_i \le x) \tag{58}$$
$$= \prod_{i=1}^{n} F_i(x), \tag{59}$$

where the third-to-last equality follows from the independence of $X_1, X_2, \dots X_n$. So the probability density function of $Y_n$ is

$$f_{Y_n}(x) = \frac{d}{dx}\prod_{i=1}^{n} F_i(x) \tag{60}$$
$$= \frac{d}{dx}F_1(x) \prod_{\substack{1\le i\le n \\ i\ne 1}} F_i(x) + \dots + \frac{d}{dx}F_n(x) \prod_{\substack{1\le i\le n \\ i\ne n}} F_i(x) \tag{61}$$
$$= f_1(x) \prod_{\substack{1\le i\le n \\ i\ne 1}} F_i(x) + \dots + f_n(x) \prod_{\substack{1\le i\le n \\ i\ne n}} F_i(x) \tag{62}$$
$$= \sum_{i=1}^{n} f_i(x) \prod_{\substack{1\le j\le n \\ j\ne i}} F_j(x). \tag{63}$$

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$\boxed{\textbf{Problem 6}}$ Consider a system with $n$ components $c_1, c_2, \dots c_n$ which are connected in series. If the component $c_i$ has failure density that is exponential with mean $\theta_i$, $i = 1, 2, \dots n$

**(a)** What is the reliability of the system? That is, find the survival function.

**(b)** What is the mean failure time of the system?

$\boxed{\textbf{Solution}}$ **(a)** Let $T_i$ represent the time to failure of component $c_i$. Then $T_i \sim \exp(\theta_i)$. The survival function for the system is

$$S(x) = 1 - F_{Y_1}(x),$$

where $Y_1 = \min(T_1, T_2, \dots, T_n)$. From Problem 5, we know that

$$F_{Y_1}(x) = 1 - \prod_{i=1}^{n}[1 - F_{T_i}(x)].$$

Since the $T_i$'s are exponentially distributed, we have

$$1 - F_{T_i}(x) = 1 - (1 - e^{-x/\theta_i}) \tag{64}$$

$$= e^{-x/\theta_i}. \tag{65}$$

Therefore

$$S(x) = 1 - \left[1 - \prod_{i=1}^{n} e^{-x/\theta_i}\right] \tag{66}$$

$$= e^{-x \sum\limits_{i=1}^{n} \frac{1}{\theta_i}}. \tag{67}$$

**(b)** From **part a**, we can see that $Y_1$ (the time to failure of the series system) has an exponential distribution with mean

$$\beta = \frac{1}{\sum\limits_{i=1}^{n} \frac{1}{\theta_i}}.$$

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**Problem 7** Suppose the components from Problem 6 are connected in parallel. Find the reliability of the system and an expression for the mean failure time.

**Solution** A parallel system fails when the last component fails. Therefore, the time to failure is $Y_n = \max(T_1, T_2, \ldots, T_n)$. Using the results of Problem 5b, the cdf of $Y_n$ is

$$F_{Y_n}(x) = \prod_{i=1}^{n} F_{T_i}(x) \tag{68}$$

$$= \prod_{i=1}^{n} [1 - e^{-x/\theta_i}]. \tag{69}$$

So the reliability of the system is

$$S(x) = 1 - F_{Y_n}(x) \tag{70}$$

$$= 1 - \prod_{i=1}^{n} [1 - e^{-x/\theta_i}]. \tag{71}$$

Letting, $T = Y_n$, the mean failure time of the system is

$$E[T] = \int_0^\infty t f_T(t)\, dt \tag{72}$$

$$= \int_0^\infty t \sum_{i=1}^{n} f_{T_i}(t) \prod_{\substack{1 \le j \le n \\ j \ne i}} F_{T_j}(t)\, dt \tag{73}$$

$$= \int_0^\infty t \sum_{i=1}^{n} \frac{1}{\theta_i} e^{-t/\theta_i} \prod_{\substack{1 \le j \le n \\ j \ne i}} [1 - e^{-t/\theta_j}]\, dt. \tag{74}$$

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$\boxed{\textbf{Problem 8}}$ Suppose $Y_1 < Y_2 < \ldots < Y_n$ are the order statistics from a random sample $X_1, X_2, \ldots, X_n$ taken from $U[\alpha, \beta]$ (i.e., a uniform distribution on the closed interval $[\alpha, \beta]$). Calculate $P(a < R < b)$, where $R = Y_n - Y_1$ is the sample range.

$\boxed{\textbf{Solution}}$ We have already determined in class that the pdf of the range $R$ for a random sample from a uniform$[\alpha, \beta]$ distribution is

$$f_R(r) = \frac{n(n-1)r^{n-2}(\beta - \alpha - r)}{(\beta - \alpha)^n},$$

for $0 \leq r \leq \beta - \alpha$. Elsewhere, it is 0. Thus,

$$\begin{aligned}
P(a < R < b) &= \int_a^b \frac{n(n-1)r^{n-2}(\beta - \alpha - r)}{(\beta - \alpha)^n}\, dr && (75)\\
&= \int_a^b \frac{n(n-1)r^{n-2}(\beta - \alpha)}{(\beta - \alpha)^n}\, dr - \int_a^b \frac{n(n-1)r^{n-1}}{(\beta - \alpha)^n}\, dr && (76)\\
&= \left[\frac{nr^{n-1}}{(\beta - \alpha)^{n-1}}\right]_a^b - \left[\frac{(n-1)r^n}{(\beta - \alpha)^n}\right]_a^b && (77)\\
&= \frac{nb^{n-1} - na^{n-1}}{(\beta - \alpha)^{n-1}} - \frac{(n-1)b^n - (n-1)a^n}{(\beta - \alpha)^n}. && (78)
\end{aligned}$$

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